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ORIGINAL PAPER

The effectiveness of *Trapezia cymodoce* in defending its host coral *Pocillopora acuta* against corallivorous *Drupella*

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Abstract

Population outbreaks of corallivorous *Drupella* gastropods have caused mass coral mortality, but there is insufficient information on their feeding behaviour to develop useful reef management strategies. This study examined the feeding rates of two *Drupella* species, *D. rugosa* (Born 1778) and *D. margariticola* (Broderip 1833), and investigated whether the presence of the coral guard crab, *Trapezia cymodoce* (Herbst 1801), could help to reduce corallivory on *Pocillopora acuta* (Lamarck 1816). Our mesocosm study showed that the feeding rate of *D. rugosa* (1.81 ± 0.95 cm^2 coral tissue/day) was significantly higher than that of *D. margariticola* (0.51 ± 0.75 cm^2 coral tissue/day). The presence of *T. cymodoce* reduced the feeding rate of *D. rugosa* by 22.9%; this was lower than that by other *Trapezia* congenerics. *Trapezia cymodoce* also did not display much aggressive behaviour towards *D. rugosa*. This study has highlighted that *D. rugosa* can cause considerable damage to corals and defence by *T. cymodoce* alone is unlikely to be effective against *Drupella* corallivory. Early detection of *Drupella* outbreaks and the subsequent development of mitigation measures remain of paramount importance in reducing the impact of corallivory on coral reefs.

Introduction

Outbreaks of the corallivorous gastropod, *Drupella*, were first recorded from the coral reefs of Southern Japan in the early 1980s (Moyer et al. 1982; Fujioka and Yamazato 1983; Awakuni 1989). Since then, densities of more than three million individuals per square kilometre (Cumming 2009) have been documented, mainly from four of the six known *Drupella* species, which are *D. cornus* (Röding 1798), *D. fragum* (Blainville 1832), *D. rugosa* (Born 1778) and *D. margariticola* (Broderip 1833) (Claremont et al. 2011). *Drupella cornus* and *D. fragum* are the best studied species, with outbreaks reported in Western Australia, Japan, Philippines and the Gulf of Aqaba (Moyer et al. 1982; Cumming 1999; Al-Horani et al. 2011); those of *D. rugosa* and *D. margariticola* were observed on reefs off Hong Kong and Thailand (Baird 1999; Morton et al. 2002; Hoeksema et al. 2013).

Sudden increases in *Drupella* populations can be detrimental to coral survival and growth. For instance, up to 75% coral mortality was reported from reefs in the Red Sea and Japan (Cumming 2009), while coral cover at Ningaloo Reef declined by up to 86% following outbreaks (Ayling and Ayling 1987; Stoddart 1989). Reefs that are impacted by coral diseases or mass bleaching are particularly vulnerable to further degradation, as high corallivory rates will impede reef recovery (Shafir et al. 2008; Bruckner et al. 2017). Predation on cultured corals and transplants can also hamper the effectiveness of reef restoration efforts (Shafir et al. 2006; Afiq-Rosli et al. 2017; Toh et al. 2017).

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Management is thus necessary but highly challenging as there is limited information on *Drupella* behaviour to effectively mitigate their impacts. The gastropods are cryptic, tending to hide in crevices during the day and emerge to feed actively at night (Moyer et al. 1982; Boucher 1986). While they can predate on non-branching corals such as massive *Porites* and plating *Montipora* (Schoepf et al. 2010), most species have a dietary preference for branching corals such as *Acropora* and *Pocillopora* (Moerland et al. 2016), as the structural complexity of these corals provides protection against predators (Boucher 1986). A female can deposit more than 3000 embryos (Sam et al. 2017) deep within coral branches, rendering the physical removal of the egg capsules impossible without causing damage to the coral host. In situ detection of embryos is also challenging, as the egg capsules are similar in colour to the substrates and thus easily overlooked in field surveys (Scott et al. 2017a).

Scleractinian corals have established mutualistic relationships with other organisms to augment their overall fitness (McKeon et al. 2012). For example, snapping shrimps (*Alpheus* spp.) can detect chemical cues from corallivores and fend off attacks by the crown-of-thorns sea star, *Acanthaster planci* (Glynn 1980). Coral guard crabs (*Trapezia* spp.), mostly found among branches of live *Pocillopora* and *Acropora* colonies (Sin and Lee 2000), are highly territorial and aggressive towards conspecifics and corallivores (Glynn 2013). They actively deter *A. planci* (Vogler et al. 2008) by clipping and pinching the soft tissues of intruders (Glynn 2013). These crabs also feed on the mucus secreted by their host corals, removing waste products, sediment, and mucus nets from the vermetid gastropods that could be deleterious to corals (Stimson 1990; Stier et al. 2010; Glynn 2013).

Reports documenting high densities of *Drupella* spp. (Tsang and Ang 2015), especially after mass bleaching events (Hoeksema et al. 2013; Bruckner et al. 2017), have increased in recent years, highlighting the importance of understanding feeding behaviour in *Drupella* spp. Most studies have focused on *D. cornus* (Turner 1994; Baird 1999; Kita et al. 2005; Shafir et al. 2008; Schoepf et al. 2010; Hoeksema et al. 2013; Tsang and Ang 2015), but knowledge of *D. rugosa* and *D. margariticola* is scant, even though their outbreaks could also pose considerable threat to reefs (Baird 1999; Hoeksema et al. 2013). Furthermore, while it is understood how commensals such as *Trapezia* spp. defend their hosts against *A. planci*, it is not apparent how successful the crabs are in reducing *Drupella* corallivory, as well as their behaviour in response to *Drupella* predation.

To investigate if *Trapezia* crabs are effective in protecting their host corals against *Drupella* spp., we first determined the feeding rates of two *Drupella* species, *D. rugosa* and *D. margariticola* and, second, evaluated the influence of *T. cymodoce,* a common coral commensal, on the feeding rate of *D. rugosa.* Third, we documented the frequency, duration, and type of behavioural responses that *T. cymodoce* exhibited in the presence of *D. rugosa* to understand the defensive mechanisms that could be employed by *Trapezia* crabs in defending their host.

Materials and methods

Sample collection

Pocillopora acuta (approximately 25 cm in maximum colony diameter), *T. cymodoce* (identified following Castro 1997a, b), *Drupella rugosa* and *D. margariticola* (both identified following Morton and Blackmore 2009) were collected from the reefs fringing Kusu (1°13.32′N, 103°51.59′E) and the seawalls at Seringat, Singapore (1°13.32′N, 103°51.33′E). All epifauna found on the corals were removed following McKeon and Moore (2014). Prior to the commencement of the study, *Pocillopora acuta* colonies were reared together with *T. cymodoce* in an outdoor tank (length × width × depth: 190 cm × 100 cm × 40 cm) with flow-through filtered seawater (mean seawater temperature 30.6 °C) at the St. John's Island National Marine Laboratory. *Drupella rugosa* and *D. margariticola* were kept in separate tanks with flow-through filtered seawater. The corals and crabs were acclimatised in the aquaria while the gastropods were starved for 7 days before the start of the experiment (McKeon and Moore 2014). There were no significant differences in the sizes of organisms used among treatments for each experiment ($P > 0.05$).

Feeding rates of *Drupella rugosa* and *D. magariticola*

Five *P. acuta* colonies were fragmented into similar sizes (3.94 ± 0.51 cm in diameter; $P > 0.05$, $n = 30$) following McKeon and Moore (2014). Each fragment was attached to 10 cm × 10 cm terracotta tiles using marine epoxy. They were each randomly placed into 30 perforated circular tanks (diameter × depth: 15 cm × 12 cm; each supplied with filtered seawater) and left to acclimatise for 2 h (McKeon and Moore 2014). Ten individuals each of *D. rugosa* and *D. margariticola* with mean shell length of 1.99 ± 0.26 and 1.95 ± 0.20 cm, respectively, were also selected.

There were three treatments: coral fragments (1) with *D. rugosa*, (2) with *D. margariticola*, and (3) without *Drupella* gastropods to function as controls and determine the basal rate of tissue loss ($n = 10$ per treatment). Photographs (Canon G7X) of two opposite sides of each coral fragment were taken underwater with scale bar every 24 h, until at least 90% of each coral fragment in the *Drupella* treatments were consumed.

The photographs were analysed using the software ImageJ (NIH) to measure the area of coral tissue lost. The

average feeding rate (cm^2/day) of each *Drupella* species was calculated by dividing the total area of coral tissue loss by the number of days (8 days in this study). Tissue loss in the control group (without gastropods) was determined using the same method. One-way ANOVA was used to test if there were any significant differences among the three treatments using square-root transformed values of feeding rates. Pairwise comparisons were subsequently performed using post hoc Tukey HSD test (SPSS Version 21.0).

Effect of *T. cymodoce* on *Drupella rugosa* corallivory rates

To determine the effect of *T. cymodoce* on *P. acuta* colonies that were affected by corallivory, only *D. rugosa* was used. *Drupella margariticola* was not studied as tissue loss from corals exposed to this gastropod was not significantly different from those in the control group (without gastropods) (refer to "Corallivory rates of *Drupella rugosa* and *D. magariticola*"). Five *P. acuta* colonies were fragmented into similar sizes (4.63 ± 0.86 cm in diameter, $P > 0.05$, $n = 16$) and attached using marine epoxy to 15 cm × 15 cm terracotta tiles. The fragments were then randomly placed into each of the 16 perforated tanks (17 cm × 17 cm × 15 cm, length × width × depth) supplied with filtered seawater and left to acclimatise for 2 h (McKeon and Moore 2014) before the start of the experiment.

There were four treatments: (1) coral fragment only (negative control to determine the basal rate of tissue loss); (2) coral fragment with one *T. cymodoce* (procedural control to check if the crab is causing coral tissue loss); (3) coral fragment with one *D. rugosa* (mean shell length 1.96 ± 0.23 cm); and (4) coral fragment with one *T. cymodoce* (mean carapace length 1.18 ± 0.13 cm) and one *D. rugosa* ($n = 16$ per treatment). *Trapezia cymodoce* were placed into each tank for 2 h to allow the crabs to acclimatise and colonise the *P. acuta* fragments. The gastropods were starved for 7 days before introducing them to the coral fragments (McKeon and Moore 2014).

Photographs were taken of the coral fragments at the start (day 1) and at the end (day 7) of the experiment. Using ImageJ, tissue loss before and after the experiment was measured (Fig. 2). One-way ANOVA with post hoc Tukey HSD test was used to test for any differences among treatments.

Interactions between *T. cymodoce* and *D. rugosa*

To examine the behavioural responses of *T. cymodoce* to *D. rugosa,* observations were conducted from dusk to dawn (1830–0630 h) following Mckeon et al. (2012), since *D. rugosa* is a nocturnal predator (Moyer et al. 1982). The interactions between *T. cymodoce* and *D. margariticola* were not studied, as tissue loss from corals exposed to this gastropod was not significantly different from those in the control group (without gastropods) (refer to "Corallivory rates of *Drupella rugosa* and *D. magariticola*"). Four treatments with ten replicates were tested: (1) a live coral fragment with one *T. cymodoce* (control); (2) a live coral fragment with one *T. cymodoce* and one *D. rugosa*; and (3) a dead coral fragment with one *T. cymodoce* and one *D. rugosa.* Each coral fragment measured approximately 5.0 cm in diameter, while the carapace length of each *T. cymodoce* was 1.23 ± 0.15 cm and the shell length of each *D. rugosa* was 1.83 ± 0.26 cm. Each treatment occupied one glass tank measuring 12 cm × 10 cm × 15 cm (length × width × depth) supplied with flow-through seawater. The coral fragments were acclimatised for 2 h after attachment to square terracotta tiles (6 cm × 6 cm). *Trapezia cymodoce* was then introduced into each tank with *P. acuta* fragments and allowed to acclimatise for another 2 h.

Eight closed circuit television cameras (Amber Indoor CCTV Camera, Singapore) were installed, such that each camera could capture activities in two treatment tanks simultaneously. The tanks were separated with opaque boards to eliminate the confounding effects of visual cues among tanks. The interactions between *T. cymodoce* and *D. rugosa* were recorded based on the ethogram adapted from Glynn (2013) (Table 1). The frequency and duration of interactions between *T. cymodoce* and *D. rugosa* were recorded following Glynn (2013), regardless of whether physical contact was made (McKeon et al. 2012). A Mann–Whitney *U* test

Table 1 Ethogram documenting the possible responses of *Trapezia cymodoce* to *Drupella rugosa* (adapted from Glynn 2013)

Actions	Definitions	Action type	
		Active	Passive
Avoidance	*Trapezia* moved to the opposite side of coral when approached		✓
Exposure	When sensing the approaching intruder, *Trapezia* moved to coral periphery and exposed itself	✓	
Flexing of chelipeds	*Trapezia* flexed its chelipeds widely when threatened	✓	
Gripping using chelae	*Trapezia* gripped onto intruder using at least one chela to prevent intruder from escaping	✓	
Gripping using walking legs	*Trapezia* gripped onto intruder using at least two opposite side walking legs	✓	
Pinching	*Trapezia* pinched part of intruder's body	✓	

was carried out to test for significant differences in the frequency and duration of interactions between *T. cymodoce* and *D. rugosa* in the presence of dead or live *P. acuta*. The most common behaviours of *T. cymodoce* when approached by *D. rugosa* were also recorded and classified as active (aggressive towards *D. rugosa*) or passive (little to no aggression towards *D. rugosa)*.

Results

Corallivory rates of *Drupella rugosa* and *D. magariticola*

After 8 days, the average tissue loss in the *D. rugosa* treatment was 9.28 ± 2.78 cm^2 ($n = 10$), higher than that in the *D. margariticola* treatment (3.27 ± 4.33 cm^2, $n = 10$) and control (1.57 ± 2.01 cm^2, $n = 10$) (Fig. 1). For both *D. rugosa* and *D. margariticola* treatments, the greatest tissue loss occurred on the first day (2.91 ± 1.82 and 0.80 ± 1.38 $cm^{2,}$ respectively). Subsequently, the rate of tissue loss in both treatments decreased over time. By the eighth day, all of the coral fragments in the *D. rugosa* treatment were at least 90% consumed when the experiment was terminated.

Average rates of daily tissue loss were significantly different among the three treatments [ANOVA, $F(2,27) = 16.906$, $P < 0.001$)] The rate of daily tissue loss for corals exposed to *D. rugosa* (1.81 ± 0.95 cm^2/day, $n = 10$) was significantly higher than that in the *D. margariticola* treatment (0.51 ± 0.75 cm^2/day, $n = 10$) ($P < 0.01$) and control (0.20 ± 0.25 cm^2/day, $n = 10$) ($P < 0.001$). There was no significant difference between the control and the *D. margariticola* treatment ($P > 0.05$).

Effects of *T. cymodoce* on *D. rugosa* corallivory

The mean daily rates of tissue loss were significantly different among treatments [ANOVA, $F(3,60) = 9.471$, $P < 0.001$]. Both treatments with *D. rugosa* had significantly higher rate of tissue loss than the two controls (Fig. 2). The negative control (only *P. acuta* fragment) had a significantly lower rate of tissue loss (0.26 ± 0.58 cm^2/day, $n = 16$) compared to treatments with coral and *D. rugosa* (1.18 ± 0.55 cm^2/day, $n = 16$; $P < 0.05$) and coral with both *D. rugosa* and *T. cymodoce* (0.91 ± 0.96 cm^2/day, $n = 16$; $P < 0.01$). The procedural control (*P. acuta* fragment with *T. cymodoce*) also had a significantly lower rate of tissue loss (0.13 ± 0.41 cm^2/day, $n = 16$) compared to treatments with coral and *D. rugosa* and coral with both *D. rugosa* and *T. cymodoce* ($P < 0.01$ and $P < 0.001$, respectively). Average tissue loss in the treatment with coral and both *T. cymodoce* and *D. rugosa* was lower than that in treatment with coral and *D. rugosa* (Fig. 2), albeit not significant ($P > 0.05$).

Behaviour of *T. cymodoce* when approached by *D. rugosa*

Interactions between *T. cymodoce* and *D. rugosa* were observed on both live and dead *P. acuta* fragments. The frequency of interactions in treatment with live *P. acuta* was significantly higher (5.4 ± 7.5, $n = 10$) than that in treatments with dead *P. acuta* (0.4 ± 0.7, $n = 10$) [$U = 14.5$, $P < 0.05$] (Table 2). The duration of interactions on live *P. acuta* was also significantly longer (7.10 ± 10.28 min, $n = 10$), compared to those on dead *P. acuta* (0.22 ± 0.44 min, $n = 10$) [$U = 18.5$, $P < 0.05$] (Table 2). For both live and dead *P. acuta* treatments, the most frequent response of *T. cymodoce* to *D. rugosa* was avoidance, i.e., *T. cymodoce* either moved or remained on the opposite side of the coral when *D. rugosa* approached the coral.

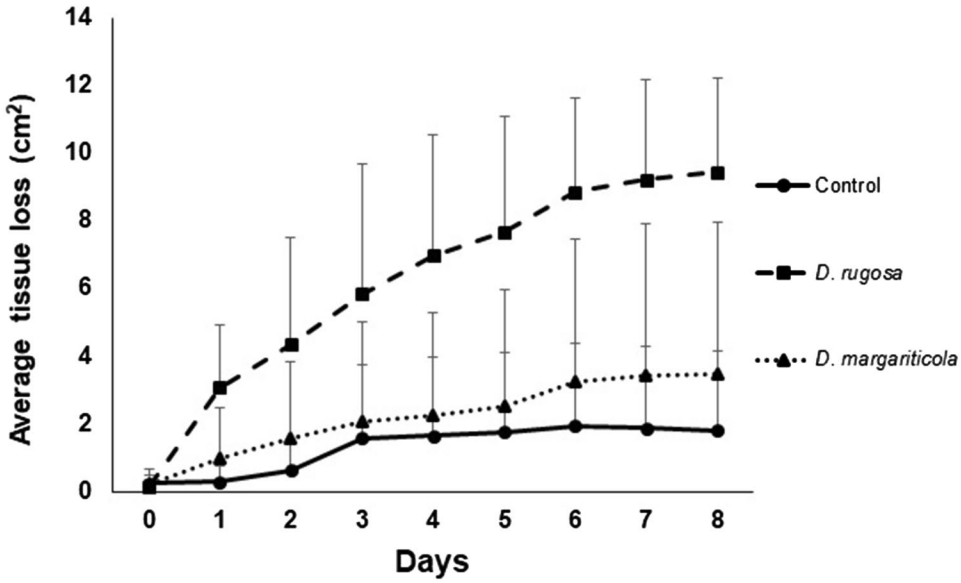

Fig. 1 Average area of tissue lost (± SD) from *Pocillopora acuta* in control group (without *Drupella*) and exposed to *Drupella rugosa* and *Drupella margariticola* ($n = 10$ per treatment)

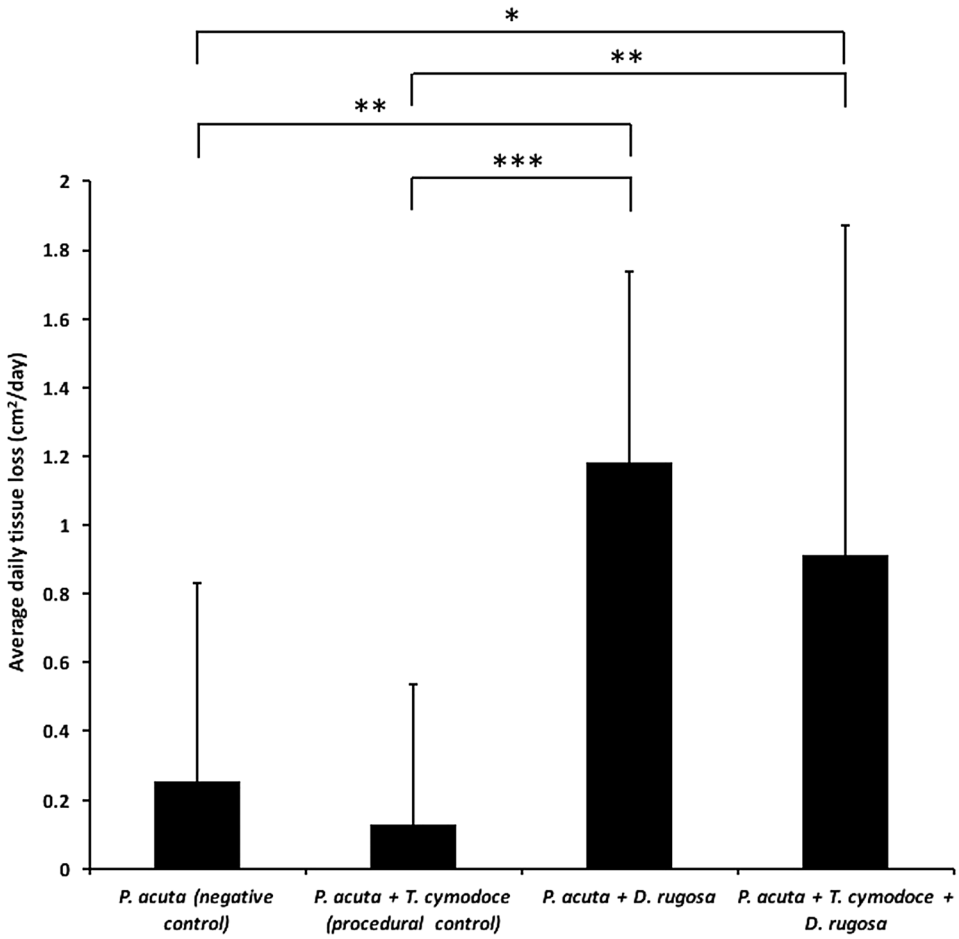

Fig. 2 Average daily tissue loss ($cm^2 \pm SD$/day) from *Pocillopora acuta* in controls (negative and procedural), in the treatment with *Drupella rugosa*, and the treatment with both *Trapezia cymodoce* and *Drupella rugosa* ($n = 16$ per treatment). Asterisks indicate statistically significant differences $*P < 0.05$, $**P < 0.01$ and $***P < 0.001$

Table 2 Frequency and duration of interactions between *Trapezia cymodoce* and *Drupella rugosa* on *Pocillopora acuta*, and the types of responses *Trapezia cymodoce* exhibited during interactions ($n = 10$ per treatment)

Treatment	Average frequency of interactions	Average duration of interactions (min)	Frequency of actions					
			Passive	Active				
			Avoidance	Gripping using walking legs	Exposure	Flexing of chelipeds	Gripping using chelae	Pinching
Live coral with crab and snail	5.4* (7.56)	7.1* (10.28)	21	19	4	3	–	1
Dead coral with crab and snail	0.44* (0.73)	0.22* (0.44)	1	–	–	–	1	–

Standard deviations are shown in parentheses. Asterisks indicate statistically significant differences ($P < 0.05$) between the treatments

Other common responses included gripping *D. rugosa* using chelipeds and walking legs, pinching the shell of *D. rugosa*, cheliped flexing, and moving out of the coral.

Discussion

High densities of corallivorous *Drupella* gastropods can cause considerable damage to coral reefs and exacerbate impacts from global climate change. Examining the corallivorous behaviour of these gastropods is imperative to enhancing management of reef habitats. Here, we show that *Drupella rugosa* can consume coral material quickly and could pose significant threats to the persistence of coral reefs. In addition, we demonstrate that *Trapezia cymodoce* is not an effective deterrent to the corallivorous gastropods.

The feeding rates of *Drupella* gastropods vary substantially among species, which could impact corals in different ways. Compared to the larger *D. cornus* that feeds at a rate of 2.6 cm^2 coral tissue/day (Rotjan and Lewis 2008),

the feeding rates of *D. rugosa* and *D. margariticola* were 1.81 cm^2 coral tissue/day and 0.51 cm^2 coral tissue/day, respectively (this study). Although *D. rugosa* is 70% the size of *D. cornus* (see Sam et al. 2017), the feeding rate to shell length ratio of the smaller muricid was higher (0.72 cm^2 coral tissue/day/cm) than that of *D. cornus* (0.65 cm^2 coral tissue/day/cm). This suggests that *D. rugosa* has the potential to cause a similar or higher extent of damage to coral reefs compared to *D. cornus*. The difference in feeding rates between *D. rugosa* (an obligate corallivore) and *D. margarticola* (an opportunistic corallivore) could also be explained by the differences in feeding apparatus of both species (Fujioka 1982, 1985; Ishida 2001; Claremont et al. 2011). *Drupella rugosa* possesses specialised mouth parts consisting of externally cuticularised proboscis and lateral teeth on the radula, which are absent in *D. margariticola* (Turner 1992; Claremont et al. 2011). This morphological specialisation protects *D. rugosa* against the nematocysts of corals and facilitates rapid feeding as compared to *D. margariticola* which may only feed on corals that had already been attacked by other *Drupella* species (Moyer et al. 1982; Turner 1992; Tsang and Ang 2015). Interestingly, outbreaks of *D. margariticola* occurred only when corals were under stress due to bleaching events or coral disease (Baker et al. 2008; Morton and Blackmore 2009). For example, an outbreak of *D. margariticola* was only reported in Hong Kong following the 1998 mass bleaching event (Cumming and McCorry 1998). The defence mechanisms of corals affected by bleaching had likely been compromised or weakened, enabling *D. margariticola* to feed even though it did not possess specialised feeding apparatus (Claremont et al. 2011).

The influence of *Trapezia* on *Drupella* differs among the species and size of the crabs (McKeon and Moore 2014). Smaller individuals of *T. punctimanus* (Odinetz 1984) and *T. serenei* (Odinetz 1983) were effective in reducing corallivory in *D. cornus* by up to 42.9% in ex situ conditions (McKeon and Moore 2014). In contrast, our study showed that the presence of *T. cymodoce* reduced daily tissue loss by 22.9%, indicating that this species was less effective in reducing corallivory by *D. rugosa* compared to other species *of Trapezia*. This could explain field observations of *Drupella* and *T. cymodoce* inhabiting the same colony (pers. obs,). While our findings suggested that the protective role of *T. cymodoce* against *Drupella* is limited, further tests (e.g., inducing conditions that are stressful to corals in controlled mesocosms) will be necessary to examine the response of *Trapezia* to opportunistic corallivores such as *D. margariticola*.

Results of the behavioural experiment underscored the ineffectiveness of *T. cymodoce* in reducing *Drupella* corallivory. *Trapezia cymodoce* displayed minimal aggression throughout the observation period, as behaviours such as exposure, flexing of chelipeds and pinching were rarely seen. Some *T. cymodoce* were observed using their walking legs to grip the gastropod shells and temporarily preventing them from feeding. In most cases, however, *T. cymodoce* predominantly avoided *D. rugosa* and some left their coral hosts temporarily. Such behaviours have been observed in the field and laboratory, as *Trapezia* migrate out of coral colonies that are overcrowded with increased competition for food and shelter (Castro 1978).

Despite attempts to pinch and grip the shell of *D. rugosa*, the chelae of *Trapezia* are more adapted for clipping slimmer and softer targets (Glynn 2013). For example, *T. ferruginea* (Latreille 1828) can pinch and clip off the flesh and thorns of *A. planci* (Castro 1997a, b; Glynn 2013). However, the chelae of *T. cymodoce* may not be effective at grasping the thick shell or soft foot of a *Drupella* gastropod. A host coral would thus have to rely on other coral ectosymbionts to stave off corallivores. For example, crustaceans that inhabit *P. acuta* such as *Alpheus lottini* and *Cymo andreossyi* are effective at defending their host, as their larger pincers are capable of crushing the shells of *Drupella* (Williams et al. 2001; Gokul and Venkataraman 2010). Hence, hosting a wide variety of ectosymbionts would increase a host coral's chances of survival against *Drupella* corallivory and detailed studies on the roles and interactions of the commensals are necessary.

Increasing frequencies of *Drupella* outbreaks will severely affect corals and coral associates. As it is challenging to effectively control *Drupella* populations, long term monitoring of coral reefs together with enhanced understanding of outbreak mechanisms and reproductive strategies in *Drupella* are important (Boucher 1986; Sam et al. 2016, 2017; Scott et al. 2017b). This will facilitate early detection of *Drupella* outbreaks and the development of necessary measures to limit or mitigate the impacts.

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Compliance with ethical standards

Ethical approval All applicable international, national, and/or institutional guidelines for the care and use of animals were followed. Permits were obtained prior to conducting the research.

Conflict of interest All authors declare that they have no conflict of interest.

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